

ASSIGNMENT #05

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, April 10th TA's Mailbox

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name in the report.
 - Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
 - Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

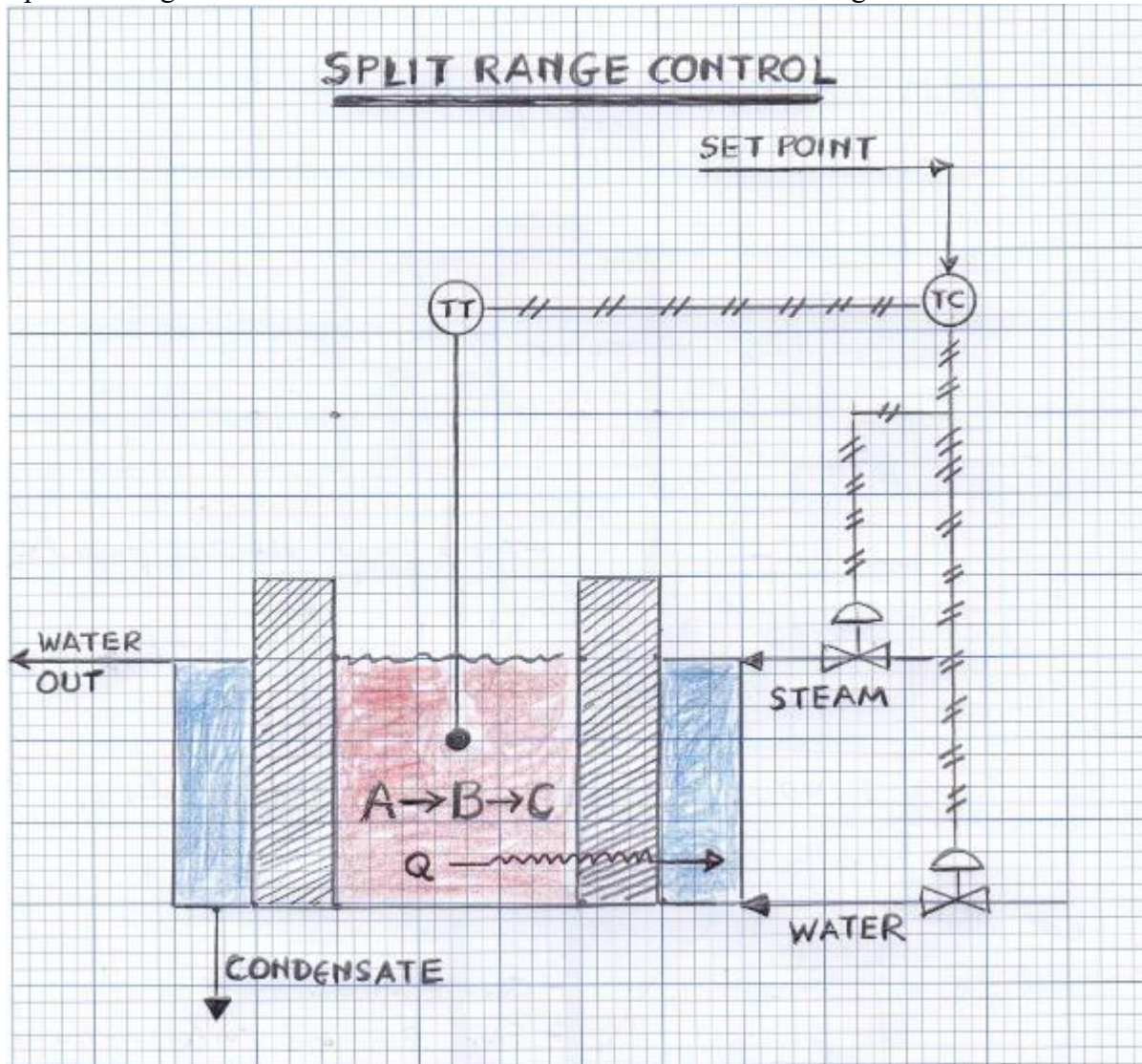
 - Assignments that are not typed will not be accepted.
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Problem A: Empirical Model Building

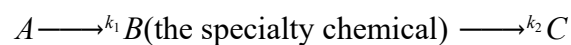
Construct the process transfer function from the multiple step test commercial data contained in the accompanying Excel file: **Assignment-05.xlsx**

Problem B: Process Control Loops and Measurement Devices for Split Range Control

A process diagram for the control of a batch reactor is shown in the Figure below:



The size of the batch reactor (red color), the metal wall thickness (black pattern) and the jacket thickness (blue color) are not drawn to scale. The objective of this assignment is to control the production of a specialty chemical compound that takes place in the batch reactor. The specialty chemical is produced by the following reactions, with the adjacent kinetic data:



$\mathfrak{R}_1 = k_{10} \exp\left\{-\frac{E_1}{RT}\right\} C_A$	$\mathfrak{R}_2 = k_{20} \exp\left\{-\frac{E_2}{RT}\right\} C_B$
$E_1 = 15,000 \text{ Btu/lbmol}$	$E_2 = 20,000 \text{ Btu/lbmol}$
$k_{10} = 729.55 \text{ min}^{-1}$	$k_{20} = 6567.6 \text{ min}^{-1}$
$\Delta H_1 = -40,000 \text{ Btu/lbmol}$	$\Delta H_2 = -50,000 \text{ Btu/lbmol}$

Batch processes offer some very distinct and challenging control problems. In a typical scenario (like the one in this assignment) the reactants are charged in the reactor which also may contain a solvent and the reactor temperature is raised by using a temperature controller that controls the flow rate of steam in the jacket of the reactor (see Figure above). Once the temperature reaches the desired point, the controller shuts down the steam valve. At this point the chemical reactions are initiated either by charging a catalyst or simply because of the high temperature. If the chemical reactions are exothermic the temperature will begin to rise. This can be detrimental to the product as well as the catalyst. To avoid a large temperature rise, the temperature controller will now open the water valve and start manipulating the water flow rate, to keep the reactor temperature at a specified set point. This is known as **split range control**.

Set points for the temperature are pre-calculated by performing optimization studies which are designed to maximize the production of the specialty chemical. Most of the time the set point varies with time and this variation is calculated by the nonlinear optimizers. In our example the set point has been pre-calculated and is given below. To control the temperature at its set point you will design a proportional controller. Your objective would be to make as much product as possible, keeping the process stable and avoiding large oscillations in the temperature and the water flow rate. Large water flow oscillations result in wear and tear of the valve. The following data are available for you to conduct your studies:

Process Variable	Reactor	Metal Wall	Reactor Jacket	Water
Initial Temperature, F °	80	80		80
Initial Concentrations, lbmol/ft ³	$C_{Ao} = 0.80$ $C_{Bo} = 0.00$ $C_{Co} = 0.00$			
Volume, ft ³	42.5	9.42	18.83	
Heat Transfer Area, ft ²	56.5 → to metal wall	56.5 → to jacket		
Heat Transfer Coeff., Btu/(hr ft F) ² °	160 → to metal wall	400 → to water 1000 → to steam		
Heat Capacity, Btu/(lb F)°	1.00	0.12		1.0
Density, lb/ft ³	50.0	512.0		62.3

Cooling water is available at 80 °F. The temperature sensor(thermocouple) has a time constant equal to $\tau_s=1$ min. The temperature transmitter sends a pneumatic signal to the temperature controller. The pneumatic signal range is [3 psi – 15 psi]. The reactor temperature range is [50 °F - 250 °F]. The water valve is linear and has been sized with $C_v=100$ gpm (psi)^{-0.5}[fully open]. The pressure drop across the valve is constant and equal to 20 psi. The valve is fully closed at a pressure signal of 9 psi and fully open at a pressure signal of 3 psi. The steam valve is also linear and has been sized to give a valve coefficient $C_v=112$ lb/min (psi)^{-0.5}[fully open]. The pressure drop across the valve is constant and equal to 0.2 psi. The valve is fully closed at a pressure signal of 9 psi and fully open at a pressure signal of 15 psi.

The controller has a bias of 7 psi and its output signal is given by

$$CO(t) = 7[\text{psi}] + K_c (SP - TT) + \frac{1}{\tau_I} \int_0^t (SP - TT) dt$$

where **TT** is the temperature transmitter signal in psi and **SP** the set point in psi. The set point has been calculated via optimization and it is given by:

$$SP = 12[\text{psi}]$$

The total time of the batch process is **60 min**. Use the 5th order Runge-Kutta method with Total Number of Solution Points=200 and Number of Steps between Solution Points=1

THE FOLLOWING TASKS ARE REQUIRED OF EACH TEAM

1. Prepare the dynamic model for this batch process. You may neglect the dynamics of the steam (but not of water) in the jacket. The reactor metal wall must also be included in the energy balances. Include a description of all symbols that you have used. The mass and energy balances must be written clearly for the three phases of the specialty chemical production, i.e.,
 - a. Heating phase using dry saturated steam from a **35 psia** steam header flowing through the reactor jacket
 - b. Cooling phase using cooling water filling the reactor jacket
 - c. Operating with jacket full of water.
2. Solve the differential equations using the **CHE Calculator** and prepare plots of the reactor temperature for controller gains $\{K_c = 2, 4, 8, 10\}$ and $\{\tau_I = \infty\}$. Give a brief description of what you are observing as you increase the controller gain. Include an Appendix in your report with the Visual Basic code of your model and the excel worksheet with your data and the plots.
3. Investigate the use of PI controller to eliminate the offset and propose a value for the controller gain and the controller rest. For these values create a plot for the reactor temperature as a function of time.

Problem A: Empirical Model Building - Solution

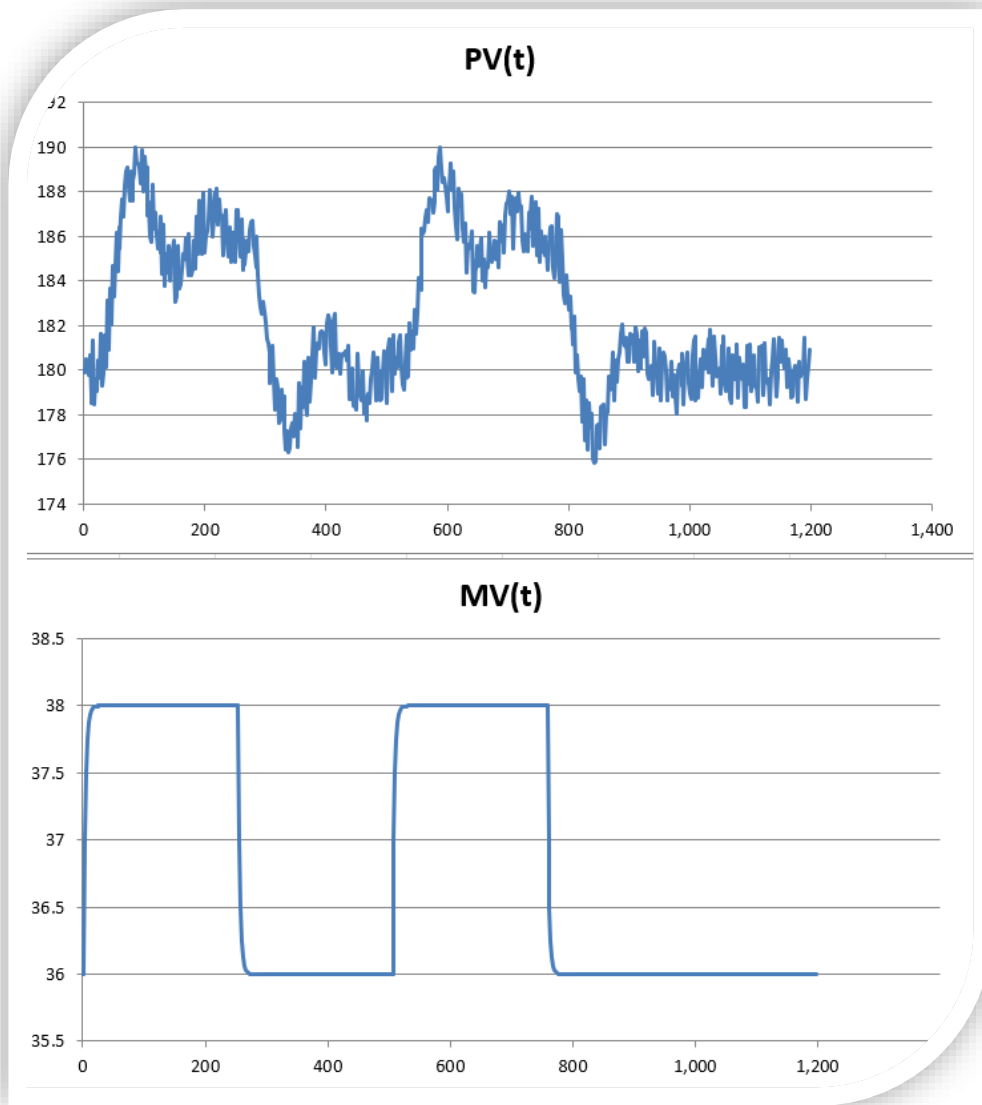


Figure 1: PV and MV graphs with correct times

Overview and objective

The purpose of this assignment is to develop an empirical process transfer function from commercial plant data obtained during multiple step tests. The available dataset includes a measured process variable $PV(t)$ and a manipulated variable $MV(t)$ that is changed in a sequence of step-like disturbances. The main objective is to approximate the process dynamics with a first-order plus dead time (FOPDT) model and to demonstrate, using the recorded time series, that the resulting model reproduces the dominant trends in the measured process response. Because the commercial data are noisy and the manipulated variable does not behave as a perfectly instantaneous step (the actuator shows brief intermediate values during transitions), parameter identification is performed using a least-squares fit rather than relying only on a few points such as 28.3% and 63.2%. The final output of this work is (1) an FOPDT model for each

step disturbance, (2) an averaged/representative set of model parameters, and (3) a full-record simulation comparing measured $PV(t)$ to the model prediction.

Data description and assumptions

The dataset contains time in seconds, the measured process variable $PV(t)$, and the manipulated variable $MV(t)$. Three major disturbances were selected for identification because they represent clear transitions between two MV plateaus and they provide sufficient data before and after each step. The step times were identified directly from the MV record.

The following assumptions were used throughout the identification:

1. **FOPDT representation is adequate for dominant behavior.** The process is approximated as a stable first-order system with an effective dead time. This model is not expected to match measurement noise or higher-order dynamics (overshoot, oscillations, secondary lags).
2. **Time sampling is uniform.** The data are sampled at a constant interval Δt (in this dataset $\Delta t = 2$ s). This permits direct discrete simulation and a simple dead-time implementation as a row shift.
3. **A single set of parameters represents the process for the full record.** After identifying parameters from three step events, the parameters are averaged to obtain one representative model used for full-record simulation. Any mismatch during different operating regions is attributed to process nonlinearity and plant disturbances not included in MV.
4. **Baseline offsets are handled using deviation variables.** The transfer function is developed in deviations about an initial baseline, then shifted back to match the measured PV scale.

Model form and step-response equation

Each disturbance is modeled using an FOPDT transfer function

$$G_p(s) = \frac{K_p e^{-\theta s}}{\tau s + 1}$$

where K_p is the steady-state gain (PV units per MV unit), τ is the first-order time constant (s), and θ is an effective dead time (s). For a single step change in MV occurring at time t_s , the corresponding time-domain response can be written as

$$PV_{\text{model}}(t) = \begin{cases} PV_0, & t < t_s + \theta \\ PV_0 + K_p \Delta MV (1 - e^{-(t-t_s-\theta)/\tau}), & t \geq t_s + \theta \end{cases}$$

where PV_0 is the average PV before the step, and ΔMV is the change in MV between its pre-step and post-step plateaus. In ideal step-test data this equation can be used with a small number of points; however, in this dataset the PV is noisy and the MV transitions include intermediate values, so parameters are best determined by minimizing the overall error across a selected window of data.

Least-squares parameter identification for each disturbance

For each disturbance, a fitting window $[t_{\text{start}}, t_{\text{end}}]$ is chosen so that it includes a stable baseline prior to the step and enough time after the step to observe the response shape. Within that window the model prediction $PV_{\text{model}}(t_i; K_p, \tau, \theta)$ is computed at each recorded time t_i . The parameters are then selected to minimize the sum of squared errors (SSE):

$$S(K_p, \tau, \theta) = \sum_{i=1}^N (PV_{\text{meas}}(t_i) - PV_{\text{model}}(t_i; K_p, \tau, \theta))^2$$

This least-squares objective uses all data points in the window and is therefore less sensitive to individual noisy samples. In practice, Excel Solver (GRG Nonlinear) was used to minimize S by changing the parameters K_p, τ, θ , while keeping the baseline values PV_0 and ΔMV fixed.

To ensure the optimization remained physically reasonable, the parameters were constrained to

$$K_p > 0, \quad \tau > 0, \quad \theta \geq 0,$$

and upper bounds were sometimes applied to prevent Solver from producing unrealistic slow dynamics when the window contained large disturbances unrelated to the MV step.

Baseline values were computed by averaging over a pre-step window on the MV plateau. The post-step plateau average was computed using a window late enough that the response was largely settled (typically well after $t_s + \theta + 3\tau$), and the window was made long to reduce noise sensitivity. It was observed during sensitivity checks that changing t_{end} and the post-plateau averaging range caused modest changes in the fitted parameters (often on the order of $\sim 10\%$). This is expected for commercial data because the PV contains noise and unmeasured disturbances; therefore, the identified model parameters should be interpreted as best-fit approximations rather than exact constants.

Summary of disturbance fits

Three step disturbances were selected from the dataset and fitted to an FOPDT model using the least-squares procedure described above. Each fit produced a gain K_p , time constant τ , and dead time θ , along with the corresponding step time and ΔMV . The fitted parameters for the three disturbances are summarized in Table 1, and the averaged values were used to form a single representative model for full-record simulation. In this dataset the parameters were reasonably consistent across the three events, so a simple average provides a defensible representative model.

Table 1: disturbance fit parameters (least squares FOPDT)						
Disturbance	Step time, t_s (s)	ΔMV	K_p	τ (s)	θ (s)	SSE
1	256	-2.0	3.23	9.70	30.22	219.48
2	508	+2.0	3.71	10.36	33.61	157.52
3	760	-2.0	3.69	12.90	28.87	182.37
Average	-	-	3.54	10.99	30.90	186.46

Using the averaged parameters from the three step disturbances, the representative FOPDT model for the process is reported as:

$$G_p(s) = \frac{\bar{K}_p e^{-\bar{\theta}s}}{\bar{\tau}s + 1} = \frac{3.54 e^{-30.9s}}{10.99s + 1}$$

where $\bar{K}_p$, $\bar{\tau}$, and $\bar{\theta}$ are obtained from Table 1.

Full-record simulation using the averaged model

After selecting representative parameters (K_p, τ, θ), the model was simulated over the entire dataset to compare the predicted response to the measured PV. Because dead time cannot be represented directly in a simple recursion without storing past values, the dead time was implemented by creating a delayed manipulated variable:

$$MV_d(t) = MV(t - \theta)$$

In discrete form, with sample time Δt , the delay corresponds to

$$n_\theta = \text{round}\left(\frac{\theta}{\Delta t}\right)$$

so the delayed MV sequence is obtained by shifting the MV column downward by n_θ rows (holding the initial MV value until sufficient history exists). This delayed MV is then used as the input to the first-order process model.

For stability and convenience, the first-order system was simulated in deviation form. Let

$$\Delta MV_d(t) = MV_d(t) - MV_0, \Delta PV(t) = PV(t) - PV_0$$

where MV_0 and PV_0 are baseline averages over an initial steady period. The first-order dynamics in deviations are

$$\tau \frac{d(\Delta PV)}{dt} + \Delta PV = K_p \Delta MV_d$$

and a discrete-time exact update for uniformly sampled data is

$$\Delta PV_k = \alpha \Delta PV_{k-1} + (1 - \alpha) K_p \Delta MV_{d,k}$$

with

$$\alpha = e^{-\Delta t/\tau}$$

Finally, the predicted PV in engineering units is reconstructed as

$$PV_{\text{model},k} = PV_0 + \Delta PV_k$$

This approach produces a smooth model prediction that responds to each MV change with the specified dead time and a first-order rise/decay. The full-record overlay plot (Figure 2) shows that the model captures the dominant shifts in PV associated with the MV pulses, while deviations remain due to measurement noise and unmodeled disturbances.

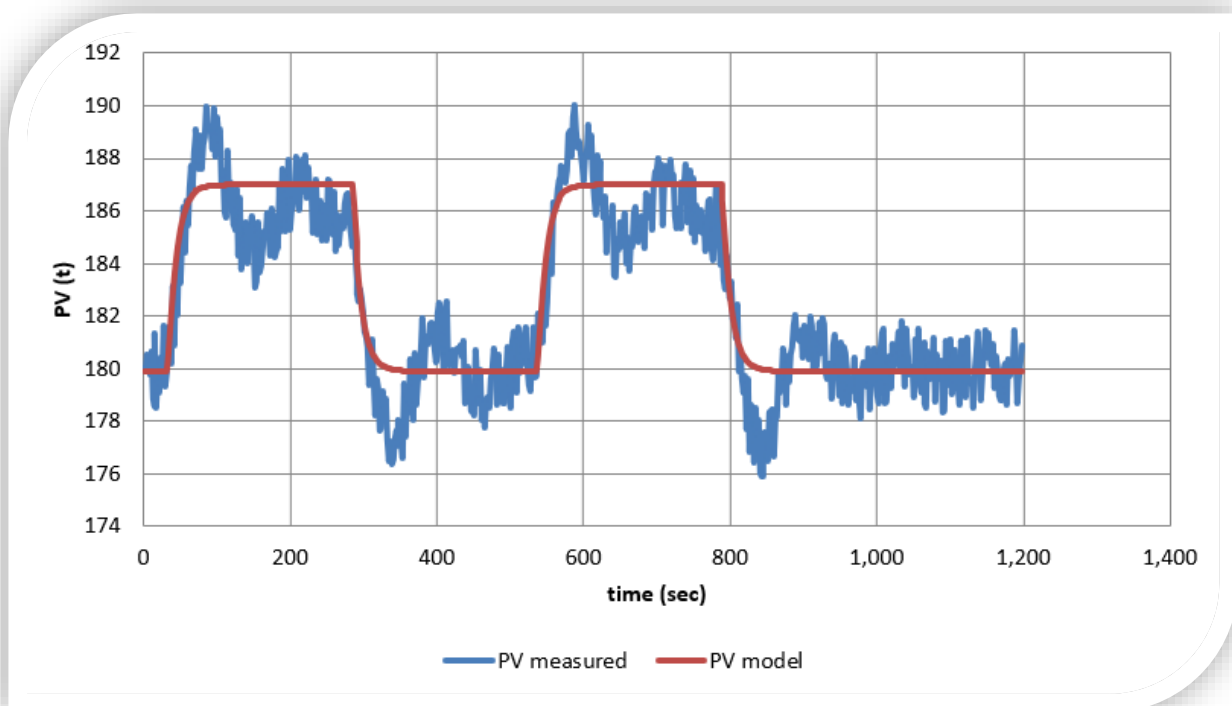


Figure 2: Process Data vs Model

Discussion of fit quality and limitations

The model-vs-data comparison indicates that the FOPDT form is appropriate for describing the dominant process behavior under these step changes. The dead time component is necessary: without it the modeled response begins immediately and does not align with the measured PV transitions. The first-order time constant provides a reasonable approximation to the rate at which PV approaches its new steady value after a step.

Despite the overall agreement, the measured PV exhibits overshoot/undershoot and strong noise during some periods, while the FOPDT model remains smooth and monotonic by construction. These discrepancies are expected because the real process likely contains additional dynamics (higher-order lags, mild oscillatory modes, or interacting disturbances) that are not represented in a first-order model. In addition, the data includes plant disturbances not captured by MV alone, so the model cannot reproduce features that are not driven by the manipulated variable.

A practical observation from the fitting process is that identified parameters depend somewhat on the chosen fitting window and the selected post-plateau averaging range. When the window end time was varied (for example, using 628 s versus 648 s in one disturbance), the resulting parameters changed on the order of $\sim 10\%$. This sensitivity is a realistic outcome in commercial data analysis and supports reporting the parameters as approximate best-fit values rather than exact constants. The final model should therefore be viewed as a compact, usable representation for control analysis and simulation, not as a full mechanistic description of the plant.

Problem B: Process Control Loops and Measurement Devices for Split Range Control - Solution

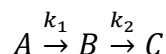
Introduction

The objective of this problem is to model and control the temperature of a batch reactor in which the desired specialty chemical, B , is produced through the consecutive reactions $A \rightarrow B \rightarrow C$. Because the second reaction consumes the desired product, temperature control is important not only for safe operation but also for maximizing selectivity toward B . The reactor is surrounded by a jacket and operated under split-range control. During the initial heat-up period, steam is used to raise the reactor temperature to the desired operating region. Once the temperature approaches the set point, the steam valve closes and the controller begins manipulating cooling water through the same output signal range. This arrangement allows one temperature controller to coordinate heating and cooling, but it also creates a piecewise dynamic problem because the jacket behavior changes as it transitions from steam service to water filling and then to operation with a full water jacket.

To answer the assignment, the process was studied in the same order required by the problem statement. First, a dynamic model was derived for the three operating phases. Next, the differential equations were solved numerically using the CHE Calculator for proportional-only control with several values of K_c , and the reactor temperature responses were compared. Finally, proportional-integral control was investigated to reduce the offset observed in the proportional-only cases and to identify a reasonable controller gain and reset time.

1. Dynamic model development

The specialty chemical is produced by two consecutive first-order reactions,



with temperature-dependent rate expressions given in Arrhenius form as

$$\begin{aligned} r_1 &= k_{10} \exp\left(-\frac{E_1}{RT_R^{abs}}\right) C_A \\ r_2 &= k_{20} \exp\left(-\frac{E_2}{RT_R^{abs}}\right) C_B \end{aligned}$$

where r_1 and r_2 are the reaction rates, k_{10} and k_{20} are pre-exponential factors, E_1 and E_2 are activation energies, R is the gas constant, and T_R^{abs} is the absolute reactor temperature in Rankine. The reactor temperature in degrees Fahrenheit is converted by

$$T_R^{abs} = T_R + 459.67$$

The symbols used in the model are defined as follows. C_A , C_B , and C_C are the concentrations of species A , B , and C in the reactor, respectively, in lbmol/ft³. T_R is the reactor temperature, T_M is the metal wall temperature, and T_J is the jacket-water temperature, all in °F. V_R , V_M , and V_J are the reactor, wall, and jacket volumes. ρ_R , ρ_M , and ρ_W are the corresponding densities, while $C_{p,R}$, $C_{p,M}$, and $C_{p,W}$ are the heat capacities. A_{RM} is the heat-transfer area between the reactor contents and the metal wall, and A_{MJ} is the area between the metal wall and the jacket. U_{RM} , U_{MW} , and U_{MS} denote the overall heat-transfer coefficients between reactor and wall, wall and water, and

wall and steam. The jacket-water mass is denoted by M_J . The temperature transmitter signal is TT in psi, the controller output is CO in psi, and the temperature set point is SP , also in psi. The controller gain is K_c , the integral reset time is τ_I , and the sensor time constant is τ_S .

The numerical values used directly from the problem statement are summarized in Table 2. The given heat-transfer coefficients were reported in $\text{Btu}/(\text{hr} \cdot \text{ft}^2 \cdot ^\circ\text{F})$, but the model was solved in minutes, so each coefficient was divided by 60 before being used in the dynamic equations. The steam saturation temperature corresponding to a 35 psia steam header was taken as approximately 259°F . This value is not listed explicitly in the assignment sheet, so it was introduced as an additional modeling constant.

Table 2: Constants used in the dynamic model		
Parameter	Value	Units
k_{10}	729.55	min^{-1}
k_{20}	6567.6	min^{-1}
E_1	15000	Btu/lbmol
E_2	20000	Btu/lbmol
ΔH_1	-40000	Btu/lbmol
ΔH_2	-50000	Btu/lbmol
R	1.987	$\text{Btu}/(\text{lbmol} \cdot \text{R})$
V_R	42.5	ft^3
V_M	9.42	ft^3
V_J	18.83	ft^3
A_{RM}	56.5	ft^2
A_{MJ}	56.5	ft^2
ρ_R	50.0	lb/ft^3
ρ_M	512.0	lb/ft^3
ρ_W	62.3	lb/ft^3
$C_{p,R}$	1.0	$\text{Btu}/(\text{lb} \cdot ^\circ\text{F})$
$C_{p,M}$	0.12	$\text{Btu}/(\text{lb} \cdot ^\circ\text{F})$
$C_{p,W}$	1.0	$\text{Btu}/(\text{lb} \cdot ^\circ\text{F})$
U_{RM}	2.667	$\text{Btu}/(\text{min} \cdot \text{ft}^2 \cdot ^\circ\text{F})$

U_{MW}	6.667	Btu(min.ft ² .°F)
U_{MS}	16.667	Btu(min.ft ² .°F)
τ_S	1	min
SP	12	psi
Bias	7	psi
$C_{v,W}$	100	lb/min.psi ^{-0.5}
ΔP_W	20	psi
$C_{v,S}$	112	lb/min.psi ^{-0.5}
ΔP_S	0.2	psi
$T_{in,W}$	80	°F
$T_{sat,35}$	259	°F

Because this is a batch reactor, there are no inlet or outlet material streams for the reacting species during the batch. Therefore, the species balances depend only on reaction. For all three phases, the reactor material balances are

$$\begin{aligned}\frac{dC_A}{dt} &= -r_1 \\ \frac{dC_B}{dt} &= r_1 - r_2 \\ \frac{dC_C}{dt} &= r_2\end{aligned}$$

The reactor energy balance is also present in all three phases. Heat is generated by the exothermic reactions and exchanged with the reactor wall, so

$$\rho_R V_R C_{p,R} \frac{dT_R}{dt} = (-\Delta H_1) V_R r_1 + (-\Delta H_2) V_R r_2 - U_{RM} A_{RM} (T_R - T_M)$$

or

$$\frac{dT_R}{dt} = \frac{(-\Delta H_1) V_R r_1 + (-\Delta H_2) V_R r_2 - U_{RM} A_{RM} (T_R - T_M)}{\rho_R V_R C_{p,R}}$$

The temperature measurement was modeled as a first-order sensor lag. Since the temperature transmitter range is 50–250°F mapped to 3–15 psi, the steady transmitter signal is

$$TT^* = 3 + \frac{12}{200} (T_R - 50)$$

which simplifies to

$$TT^* = 0.06T_R$$

and the transmitter dynamics are

$$\frac{dTT}{dt} = \frac{TT^* - TT}{\tau_S}$$

The controller output was given in the assignment as

$$CO(t) = 7 + K_c(SP - TT) + \frac{1}{\tau_I} \int_0^t (SP - TT) dt$$

For proportional-only control, the integral term is removed by letting $\tau_I \rightarrow \infty$, which makes $1/\tau_I = 0$. For PI control, an additional state variable was introduced,

$$I_E = \int_0^t (SP - TT) dt$$

so that

$$\frac{dI_E}{dt} = SP - TT$$

and

$$CO = 7 + K_c(SP - TT) + \frac{1}{\tau_I} I_E$$

The split-range valve relations determine whether steam or cooling water is active. For the steam valve, which is closed at 9 psi and fully open at 15 psi,

$$x_S = \begin{cases} 0, & CO \leq 9 \\ \frac{CO - 9}{6}, & 9 < CO < 15 \\ 1, & CO \geq 15 \end{cases}$$

For the water valve, which is fully open at 3 psi and closed at 9 psi,

$$x_W = \begin{cases} 1, & CO \leq 3 \\ \frac{9 - CO}{6}, & 3 < CO < 9 \\ 0, & CO \geq 9 \end{cases}$$

The cooling-water flow rate is

$$Q_W = C_{v,W} x_W \sqrt{\Delta P_W}$$

and the corresponding water mass flow rate is

$$W_W = Q_W \left(\frac{\rho_W}{7.4805} \right)$$

where the factor 7.4805 converts ft³ to gallons.

Because the process passes through three physically different operating periods, the wall and jacket balances change from phase to phase.

Phase A: Heating with dry saturated steam

In the heating phase, the jacket contains steam and its dynamics are neglected as instructed. Therefore, no jacket-water accumulation is modeled, and the wall receives heat from steam condensation. The wall energy balance is

$$\rho_M V_M C_{p,M} \frac{dT_M}{dt} = U_{RM} A_{RM} (T_R - T_M) + x_S U_{MS} A_{MJ} (T_{sat,35} - T_M)$$

During this phase, no water is present in the jacket, so

$$\frac{dM_J}{dt} = 0, \frac{dT_J}{dt} = 0$$

Phase B: Cooling with water filling the jacket

Once the controller output falls into the cooling range, the steam valve closes and the water valve opens. In this phase the jacket is filling, so the jacket-water mass is dynamic:

$$\frac{dM_J}{dt} = W_W$$

The wall now transfers heat to the water rather than to steam:

$$Q_{MW} = U_{MW} A_{MJ} (T_M - T_J)$$

and the wall balance becomes

$$\rho_M V_M C_{p,M} \frac{dT_M}{dt} = U_{RM} A_{RM} (T_R - T_M) - U_{MW} A_{MJ} (T_M - T_J)$$

The jacket-water energy balance is

$$M_J C_{p,W} \frac{dT_J}{dt} = W_W C_{p,W} (T_{in,W} - T_J) + U_{MW} A_{MJ} (T_M - T_J)$$

This phase is numerically sensitive because the derivative of T_J is inversely proportional to M_J . Starting the jacket mass too close to zero caused instability in the calculations, so a small finite initial jacket-water inventory was assigned at the beginning of Phase B to obtain stable integration.

Phase C: Jacket full of water

Once the jacket is completely full, the water holdup remains constant at

$$M_{J,\text{full}} = \rho_W V_J = 1173.109 \text{ lb}$$

Therefore,

$$\frac{dM_J}{dt} = 0$$

The wall balance remains the same as in Phase B, while the jacket-water energy balance becomes

$$M_{J,\text{full}} C_{p,W} \frac{dT_J}{dt} = W_W C_{p,W} (T_{in,W} - T_J) + U_{MW} A_{MJ} (T_M - T_J)$$

This completes the dynamic model required in Part 1. The equations were solved in three separate numerical phases because the physics of the jacket changes abruptly from steam heating, to water filling, to full-jacket circulation.

2. Numerical solution and proportional-control study

After deriving the model, the equations were solved in the CHE Calculator using the fifth-order Runge-Kutta method. The total integration time for each phase was selected large enough to capture the transition, and the final row of one phase was used to initialize the next. The phases were then stitched together using shifted time coordinates so that a single continuous reactor-temperature plot could be prepared for each controller gain. The detailed code is explained and shown in the Appendix.

The initial conditions for Phase A were those stated in the problem: $C_{A0} = 0.80$, $C_{B0} = 0$, $C_{C0} = 0$, and $T_R = T_M = T_J = 80^\circ\text{F}$. The initial transmitter signal was calculated from the temperature-transmitter mapping, which gave $TT_0 = 4.8$ psi. For the proportional-only study, the integral term was disabled by setting $1/\tau_I = 0$. The proportional gains studied were $K_c = \{2, 4, 8, 10\}$.

To determine the handoff between phases, the controller output and valve positions were monitored. Phase A ended at the first row where the water valve began to open, or equivalently the first row where $CO < 9$ psi. Phase B ended at the first row where the jacket-water mass reached the full-jacket value of 1173.109 lb. Phase C then began from that state, with the jacket mass fixed at its full value.

This phase-by-phase approach was necessary because solving the full split-range system as a single continuous ODE created numerical instability near the transition from steam heating to water filling. The problem arose because the jacket-water energy balance in Phase B contains M_J in the denominator. When M_J is extremely small, even moderate heat transfer from the wall causes unrealistically large derivatives in jacket-water temperature. Using three separate solution stages, with a finite initial water mass in Phase B, eliminated this difficulty and produced physically reasonable results.

The proportional-control responses are shown in Figure 3. As the proportional gain increased, the reactor temperature response became slightly more aggressive. The first peak became higher, the cooling action began somewhat earlier, and the oscillations that followed the phase transitions became more pronounced. This behavior is consistent with proportional control theory: a larger gain produces a larger change in controller output for the same temperature error, which causes both the steam shutoff and water-valve response to become more forceful.

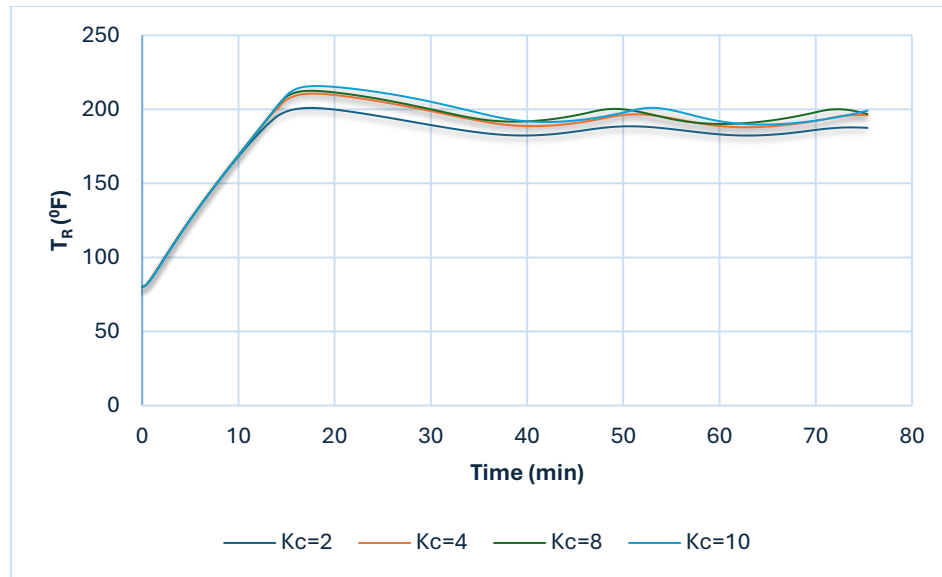


Figure 3: K_c Comparison on Offset

At the same time, the proportional-only curves all displayed a remaining offset. The set point of 12 psi corresponds to a reactor-temperature target of approximately 200°F. However, after the major transient died out, the reactor temperature did not remain exactly at 200°F. Instead, it settled slightly below or around that value depending on the gain. This remaining difference between the sustained reactor temperature and the set point is the proportional offset. It occurs because proportional-only control requires a nonzero error to maintain a nonzero corrective action. In other words, some deviation from the set point must remain in order for the controller to continue generating the output needed to balance heat generation and heat removal.

Among the proportional-only cases, $K_c = 2$ gave the smoothest response but also showed the largest remaining offset. Increasing K_c from 2 to 4, 8, and 10 reduced the offset somewhat, but the improvements came with more overshoot and somewhat greater oscillatory behavior. The higher-gain cases were still stable, but they were clearly more aggressive than the lower-gain case. Therefore, the proportional-only study demonstrated the standard tradeoff: stronger proportional action reduces offset but worsens the transient response.

3. PI control study and selection of controller settings

Because the proportional-only controller retained offset, a proportional-integral controller was next investigated. In this part, the integral state

$$I_E = \int_0^t (SP - TT) dt$$

was activated, and the controller output became

$$CO = 7 + K_c(SP - TT) + \frac{1}{\tau_I} I_E$$

The purpose of the integral term is to accumulate error over time and continue adjusting the controller output until the sustained error is driven much closer to zero. Thus, if the proportional-

only controller settles below the set point, the integral term keeps increasing the output correction until that offset is reduced.

The PI study was performed in two steps. First, K_c was held at 2 and a reset-time analysis was carried out for $\tau_I = 20, 30, 40$, and 50 min. The resulting responses are shown in Figure 4. As expected, smaller values of τ_I produced stronger integral action, because the coefficient $1/\tau_I$ became larger. The $\tau_I = 20$ min case was clearly too aggressive. It produced the largest overshoot and the least desirable transient behavior. Increasing τ_I weakened the integral action and made the response smoother. The $\tau_I = 50$ min case was smooth, but it also appeared somewhat sluggish. The best balance between offset removal and acceptable transient performance appeared to occur near $\tau_I = 40$ min. For this reason, $\tau_I = 40$ min was selected for a second comparison in which K_c was varied.

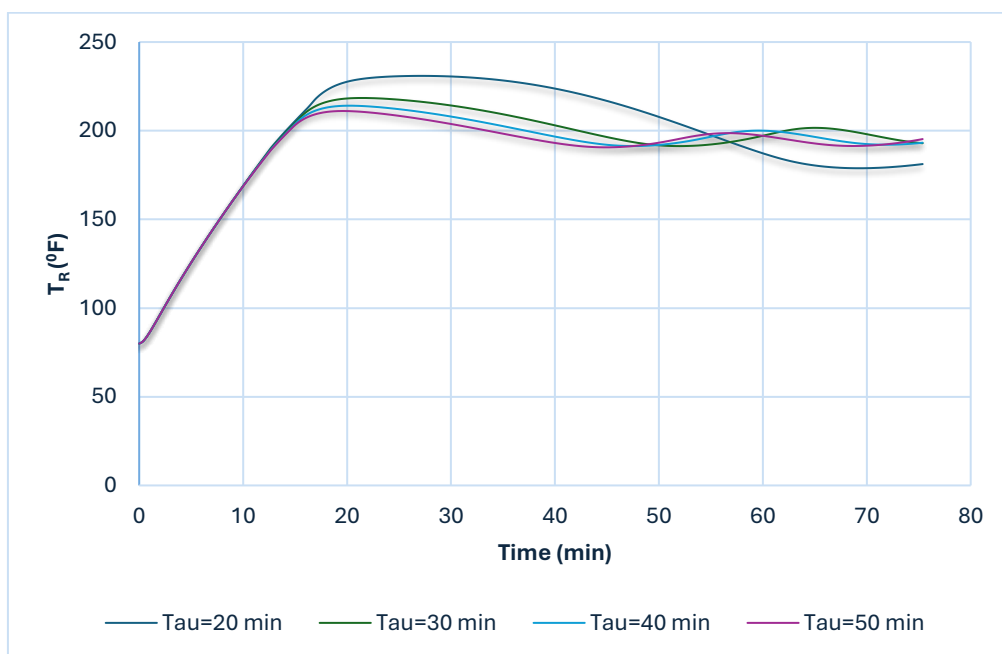


Figure 4: τ_I Analysis for $K_c = 2$

Next, with $\tau_I = 40$ min fixed, the controller gain was tested again for $K_c = 2, 4, 8$, and 10. The results are shown in Figure 5. With integral action present, the curves moved closer to the 200°F target than the proportional-only results, indicating that the offset had been reduced. However, the same tradeoff with gain remained. The larger gains, especially $K_c = 8$ and $K_c = 10$, gave sharper overshoot and more oscillatory behavior. The smallest gain, $K_c = 2$, was stable but still somewhat weaker in correcting the offset. Based on the overall shape of the response, $K_c = 4$ provided the best compromise. It reduced the offset more effectively than $K_c = 2$, while avoiding the stronger oscillation seen at $K_c = 8$ and $K_c = 10$.

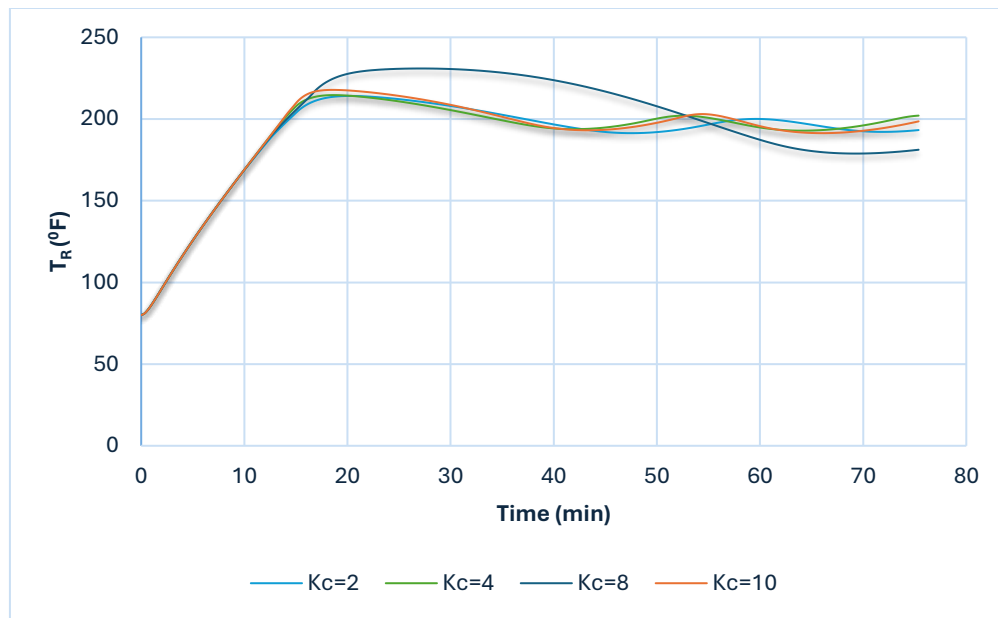


Figure 5: K_c Analysis for $\tau_I = 40$ min

Therefore, the recommended PI tuning from this study is

$$K_c = 4, \tau_I = 40 \text{ min}$$

This tuning produced a smooth response, reduced the offset observed in the proportional-only controller, and avoided the excessive aggressiveness associated with smaller τ_I or larger K_c .

Conclusion

A dynamic model was derived for a batch reactor under split-range temperature control by writing material balances for the reacting species and energy balances for the reactor, metal wall, and jacket. Because the jacket physics change substantially as the process moves from steam heating to cooling-water fill and then to full-jacket circulation, the model was solved as three consecutive phases. This approach produced stable numerical integration and captured the main behavior of the process.

The proportional-only study showed that increasing controller gain reduced the offset somewhat but also increased overshoot and oscillation. This confirmed that proportional control alone cannot fully eliminate steady-state deviation from the set point. Adding integral action reduced the offset as expected, but aggressive reset times produced poor transient behavior. Among the PI cases studied, $\tau_I = 40$ min gave the best reset-time behavior, and $K_c = 4$ gave the best overall compromise between offset removal and smoothness. Therefore, the selected controller tuning for this process was $K_c = 4$ and $\tau_I = 40$ min.

Appendix I: ODE Code

```
Function fODE_A(T As Double, u() As Double, Optional Param As Range)

    Dim n As Integer
    n = UBound(u)
    ReDim f(1 To n) As Double

    ' Read all model and controller parameters from the worksheet
    Dim k10 As Double, k20 As Double, E1 As Double, E2 As Double
    Dim dH1 As Double, dH2 As Double, Rgas As Double
    Dim VR As Double, VM As Double, ARM As Double, AMJ As Double
    Dim rhoR As Double, rhoM As Double
    Dim CpR As Double, CpM As Double
    Dim URM As Double, UMS As Double
    Dim tauS As Double, SP As Double, bias As Double
    Dim Kc As Double, invTauI As Double
    Dim Tsat35 As Double

    k10 = Param.Cells(1, 1)
    k20 = Param.Cells(2, 1)
    E1 = Param.Cells(3, 1)
    E2 = Param.Cells(4, 1)
    dH1 = Param.Cells(5, 1)
    dH2 = Param.Cells(6, 1)
    Rgas = Param.Cells(7, 1)
    VR = Param.Cells(8, 1)
    VM = Param.Cells(9, 1)
    ARM = Param.Cells(11, 1)
    AMJ = Param.Cells(12, 1)
    rhoR = Param.Cells(13, 1)
    rhoM = Param.Cells(14, 1)
    CpR = Param.Cells(16, 1)
    CpM = Param.Cells(17, 1)
    URM = Param.Cells(19, 1)
    UMS = Param.Cells(21, 1)
    tauS = Param.Cells(22, 1)
    SP = Param.Cells(23, 1)
    bias = Param.Cells(24, 1)
    Kc = Param.Cells(25, 1)
    invTauI = Param.Cells(26, 1)
    Tsat35 = Param.Cells(32, 1)

    ' Unpack the state variables
    Dim CA As Double, CB As Double, CC As Double
    Dim TR As Double, TM As Double, TT As Double
    Dim MJ As Double, TJ As Double, IE As Double

    CA = u(1)
    CB = u(2)
    CC = u(3)
    TR = u(4)
    TM = u(5)
    TT = u(6)
    MJ = u(7)
    TJ = u(8)
    IE = u(9)

    ' Prevent concentrations from becoming slightly negative during integration
    If CA < 0# Then CA = 0#
    If CB < 0# Then CB = 0#
    If CC < 0# Then CC = 0#
```

```

' Calculate absolute temperature and reaction rates
Dim TRabs As Double
Dim r1 As Double, r2 As Double
Dim TTss As Double, err As Double
Dim CO As Double, xs As Double
Dim Qrxn As Double, QRM As Double, QMS As Double

TRabs = TR + 459.67
If TRabs < 1# Then TRabs = 1#

r1 = k10 * Exp(-E1 / (Rgas * TRabs)) * CA
r2 = k20 * Exp(-E2 / (Rgas * TRabs)) * CB

' Evaluate the transmitter and controller equations
TTss = 3# + (12# / 200#) * (TR - 50#)

err = SP - TT
CO = bias + Kc * err + invTauI * IE
If CO < 3# Then CO = 3#
If CO > 15# Then CO = 15#

' Determine steam-valve opening for split-range control
If CO <= 9# Then
    xs = 0#
Else
    xs = (CO - 9#) / 6#
End If
If xs < 0# Then xs = 0#
If xs > 1# Then xs = 1#

' Compute reaction heat, reactor-to-wall heat transfer, and steam heating
Qrxn = (-dH1) * VR * r1 + (-dH2) * VR * r2
QRM = URM * ARM * (TR - TM)
QMS = xs * UMS * AMJ * (Tsats35 - TM)

' Material balances on A, B, and C
f(1) = -r1
f(2) = r1 - r2
f(3) = r2

' Reactor and wall energy balances for the steam-heating phase
f(4) = (Qrxn - QRM) / (rhoR * VR * CpR)
f(5) = (QRM + QMS) / (rhoM * VM * CpM)

' Transmitter dynamics
f(6) = (TTss - TT) / tauS

' In Phase A, jacket-water mass and jacket-water temperature do not change
f(7) = 0#
f(8) = 0#

' Integral-of-error state
f(9) = err

fODE_A = f
End Function

```

```

Function fODE_B(T As Double, u() As Double, Optional Param As Range)

    Dim n As Integer
    n = UBound(u)
    ReDim f(1 To n) As Double

    ' Read all model and controller parameters from the worksheet
    Dim k10 As Double, k20 As Double, E1 As Double, E2 As Double
    Dim dH1 As Double, dH2 As Double, Rgas As Double
    Dim VR As Double, VM As Double, ARM As Double, AMJ As Double
    Dim rhoR As Double, rhoM As Double, rhoW As Double
    Dim CpR As Double, CpM As Double, CpW As Double
    Dim URM As Double, UMW As Double
    Dim tauS As Double, SP As Double, bias As Double
    Dim Kc As Double, invTauI As Double
    Dim CvW As Double, dPW As Double, TinW As Double

    k10 = Param.Cells(1, 1)
    k20 = Param.Cells(2, 1)
    E1 = Param.Cells(3, 1)
    E2 = Param.Cells(4, 1)
    dH1 = Param.Cells(5, 1)
    dH2 = Param.Cells(6, 1)
    Rgas = Param.Cells(7, 1)
    VR = Param.Cells(8, 1)
    VM = Param.Cells(9, 1)
    ARM = Param.Cells(11, 1)
    AMJ = Param.Cells(12, 1)
    rhoR = Param.Cells(13, 1)
    rhoM = Param.Cells(14, 1)
    rhoW = Param.Cells(15, 1)
    CpR = Param.Cells(16, 1)
    CpM = Param.Cells(17, 1)
    CpW = Param.Cells(18, 1)
    URM = Param.Cells(19, 1)
    UMW = Param.Cells(20, 1)
    tauS = Param.Cells(22, 1)
    SP = Param.Cells(23, 1)
    bias = Param.Cells(24, 1)
    Kc = Param.Cells(25, 1)
    invTauI = Param.Cells(26, 1)
    CvW = Param.Cells(27, 1)
    dPW = Param.Cells(28, 1)
    TinW = Param.Cells(31, 1)

    ' Unpack the state variables
    Dim CA As Double, CB As Double, CC As Double
    Dim TR As Double, TM As Double, TT As Double
    Dim MJ As Double, TJ As Double, IE As Double

    CA = u(1)
    CB = u(2)
    CC = u(3)
    TR = u(4)
    TM = u(5)
    TT = u(6)
    MJ = u(7)
    TJ = u(8)
    IE = u(9)

    ' Prevent concentrations and jacket mass from becoming unrealistically small
    If CA < 0# Then CA = 0#
    If CB < 0# Then CB = 0#
    If CC < 0# Then CC = 0#
    If MJ < 35# Then MJ = 35#

    ' Calculate absolute temperature and reaction rates
    Dim TRabs As Double
    Dim r1 As Double, r2 As Double
    Dim TTss As Double, err As Double
    Dim CO As Double, xw As Double
    Dim Ww As Double
    Dim Qrxn As Double, QRM As Double, QMW As Double

    TRabs = TR + 459.67
    If TRabs < 1# Then TRabs = 1#

    r1 = k10 * Exp(-E1 / (Rgas * TRabs)) * CA
    r2 = k20 * Exp(-E2 / (Rgas * TRabs)) * CB

    ' Evaluate the transmitter and controller equations
    TTss = 3# + (12# / 200#) * (TR - 50#)

```

```

err = SP - TT
CO = bias + Kc * err + invTauI * IE
If CO < 3# Then CO = 3#
If CO > 15# Then CO = 15#

' Changed for Phase B: steam heating is removed, cooling-water flow is activated,
' jacket-water mass now increases, and jacket-water temperature becomes dynamic
If CO <= 3# Then
    xw = 1#
ElseIf CO < 9# Then
    xw = (9# - CO) / 6#
Else
    xw = 0#
End If

Ww = CvW * xw * Sqr(dFW) * (rhoW / 7.4805)

Qrxn = (-dH1) * VR * r1 + (-dH2) * VR * r2
QRM = URM * ARM * (TR - TM)
QMW = UMW * AMJ * (TM - TJ)

' Material balances on A, B, and C
f(1) = -r1
f(2) = r1 - r2
f(3) = r2

' Reactor and wall energy balances for the jacket-filling phase
f(4) = (Qrxn - QRM) / (rhoR * VR * CpR)
f(5) = (QRM - QMW) / (rhoM * VM * CpM)

' Transmitter dynamics
f(6) = (TTss - TT) / tauS

' Jacket-water mass and energy balances during filling
f(7) = Ww
f(8) = (Ww * CpW * (TinW - TJ) + QMW) / (MJ * CpW)

' Integral-of-error state
f(9) = err

fODE_B = f

End Function

```

```

Function fODE_C(T As Double, u() As Double, Optional Param As Range)

    Dim n As Integer
    n = UBound(u)
    ReDim f(1 To n) As Double

    ' Read all model and controller parameters from the worksheet
    Dim k10 As Double, k20 As Double, E1 As Double, E2 As Double
    Dim dH1 As Double, dH2 As Double, Rgas As Double
    Dim VR As Double, VM As Double, VJ As Double, ARM As Double, AMJ As Double
    Dim rhoR As Double, rhoM As Double, rhoW As Double
    Dim CpR As Double, CpM As Double, CpW As Double
    Dim URM As Double, UMW As Double
    Dim tauS As Double, SP As Double, bias As Double
    Dim Kc As Double, invTauI As Double
    Dim CvW As Double, dFW As Double, TinW As Double

    k10 = Param.Cells(1, 1)
    k20 = Param.Cells(2, 1)
    E1 = Param.Cells(3, 1)
    E2 = Param.Cells(4, 1)
    dH1 = Param.Cells(5, 1)
    dH2 = Param.Cells(6, 1)
    Rgas = Param.Cells(7, 1)
    VR = Param.Cells(8, 1)
    VM = Param.Cells(9, 1)
    VJ = Param.Cells(10, 1)
    ARM = Param.Cells(11, 1)
    AMJ = Param.Cells(12, 1)
    rhoR = Param.Cells(13, 1)
    rhoM = Param.Cells(14, 1)
    rhoW = Param.Cells(15, 1)
    CpR = Param.Cells(16, 1)
    CpM = Param.Cells(17, 1)
    CpW = Param.Cells(18, 1)
    URM = Param.Cells(19, 1)
    UMW = Param.Cells(20, 1)
    tauS = Param.Cells(22, 1)
    SP = Param.Cells(23, 1)
    bias = Param.Cells(24, 1)

```

```

Kc = Param.Cells(25, 1)
invTauI = Param.Cells(26, 1)
CvW = Param.Cells(27, 1)
dPW = Param.Cells(28, 1)
TinW = Param.Cells(31, 1)

' Unpack the state variables
Dim CA As Double, CB As Double, CC As Double
Dim TR As Double, TM As Double, TT As Double
Dim MJ As Double, TJ As Double, IE As Double

CA = u(1)
CB = u(2)
CC = u(3)
TR = u(4)
TM = u(5)
TT = u(6)
MJ = u(7)
TJ = u(8)
IE = u(9)

' Prevent concentrations from becoming slightly negative during integration
If CA < 0# Then CA = 0#
If CB < 0# Then CB = 0#
If CC < 0# Then CC = 0#

' Calculate absolute temperature and reaction rates
Dim TRabs As Double
Dim r1 As Double, r2 As Double
Dim TTss As Double, err As Double
Dim CO As Double, xw As Double
Dim Ww As Double
Dim Qrxn As Double, QRM As Double, QMW As Double
Dim Mfull As Double

TRabs = TR + 459.67
If TRabs < 1# Then TRabs = 1#

r1 = k10 * Exp(-E1 / (Rgas * TRabs)) * CA
r2 = k20 * Exp(-E2 / (Rgas * TRabs)) * CB

' Evaluate the transmitter and controller equations
TTss = 3# + (12# / 200#) * (TR - 50#)

err = SP - TT
CO = bias + Kc * err + invTauI * IE
If CO < 3# Then CO = 3#
If CO > 15# Then CO = 15#

' Changed for Phase C: the jacket is now full, so jacket mass is fixed
' and the jacket-water energy balance uses the constant full-jacket mass
If CO <= 3# Then
    xw = 1#
ElseIf CO < 9# Then
    xw = (9# - CO) / 6#
Else
    xw = 0#
End If

Ww = CvW * xw * Sqr(dPW) * (rhoW / 7.4805)

Qrxn = (-dH1) * VR * r1 + (-dH2) * VR * r2
QRM = URM * ARM * (TR - TM)
QMW = UMW * AMJ * (TM - TJ)

Mfull = rhoW * VJ

' Material balances on A, B, and C
f(1) = -r1
f(2) = r1 - r2
f(3) = r2

' Reactor and wall energy balances for the full-jacket phase
f(4) = (Qrxn - QRM) / (rhoR * VR * CpR)
f(5) = (QRM - QMW) / (rhoM * VM * CpM)

' Transmitter dynamics
f(6) = (TTss - TT) / tauS

' Jacket mass is fixed, and jacket-water temperature changes using the full-jacket mass
f(7) = 0#
f(8) = (Ww * CpW * (TinW - TJ) + QMW) / (Mfull * CpW)

' Integral-of-error state
f(9) = err

fODE_C = f

End Function

```

Appendix II: References

- [1] Caracostas, Michael. *CHE Calculator, Version 13*. Excel-based chemical engineering calculation and dynamic modeling tool. University of Illinois Chicago, Department of Chemical Engineering, 2026.
- [2] Caracoitas, Michael. *Introduction to Chemical Process Control*. Chapter 5. University of Illinois Chicago, Department of Chemical Engineering, 2026.
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- [4] OpenAI. *ChatGPT* (GPT-based large language model). San Francisco: OpenAI, accessed March 2026. <https://chat.openai.com>.